

“INTEROPERABILITY Is REQUIRED For Devices, Networks And Applications”

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As we witness the Internet of Things (IoT) slowly become the new norm, efforts towards cementing IoT standards by Indian standardisation bodies have been enormous. Sushil Kumar, deputy director general - IoT, Telecommunication Engineering Center (TEC) reveals to Baishakhi Dutta how IoT designers can refer to TEC's technical resources for creating a standardised IoT ecosystem

Q. Please provide an overview of the growing IoT landscape in India.

A. Machine-to-machine (M2M) communication and the IoT are the most disrupted technologies today. These have several associated technologies, such as artificial intelligence (AI), Big Data, machine learning and 5G, among others.

Globally, by 2020, there will be about 25 billion connected devices. As per our estimations, there will be around 2.6 billion connected devices in India.

Recently, National Digital Communication Policy released by Department of Telecom (DoT) projected five billion devices to be connected by 2022. For this, the ecosystem is required.

The IoT will play a huge role in smart cities for their development and such projects as smart metering, smart grid and connected vehicles (whose standards have been released by Ministry of Road Transport & Highways). With more such projects, number of connected devices will also increase in India.

Q. How is TEC working to solve the challenges in the IoT ecosystem?

A. TEC, a standardisation and advisory body of DoT, is working on various points raised by Aruna Sundarajan, secretary, DoT.

DoT has released a 13-digit number scheme for M2M communication, where gateways will be directly connected to the ESDM or PLMN network (which will work on SIM-based devices). The scheme will coexist with the 10-digit number system (for voice calling). With this technology, for example, a phone will also work as a device for gateways.

Another example is a hospital that is short on beds. With the IoT, after a surgery, the patient can be shifted to home and from there data can be transmitted to the hospital. So, the hospital bed can be given to another patient.

Interoperability is required for devices, networks and applications. Without it, large-scale production is not possible and, hence, prices cannot come down. Communication is important for this.

Q. How can IoT solution providers benefit from embedded SIM applications?

A. The GSMA guideline document was released in 2015 for the embedded SIM. Ministry of Road Transport & Highways and Bureau of Indian Standards (BIS) have included it in their standards for vehicle-tracking devices. The guidelines have been made essential for all passenger-commercial vehicles like cabs.

If you place the regular SIM in the vehicle, its life will not be more than four years. However, the embedded SIM, which is in the form of an IC, cannot be taken out and thrown, and its life will cover the entire life of the vehicle.

The SIM is useful for remote asset monitoring for smart ambulances, spectrum for low-power wireless technologies like LoRa and Sigfox and horizontal service layers.

Big Data will play a big role in making various verticals smart using the IoT. It will help create intelligence, which can be used for operational and planning activities. We are working with ITU-T study group 2020 on the IoT and its application in smart cities.

Q. Which type of communication technology should be used for which application area?

A. LoRa works in T-licence frequency band and can be transmitted to large distances. In rural areas it can transmit up to 10km or even 12km, in cities up to 2.5km and in cellular domain using SIM technology it can cover many metres. For example, if cellular signal does not reach the basement of an apartment, it can use an indoor base transceiver station (BTS).

LoRa signals can also reach such areas, but for that the BTS should be removed. This adds to the cost. So, technologies like RF Mesh and PLC can take metering to the transformer level, aggregate data (according to number of flats in a society) and transmit it on optical fibres or through a mobile SIM (3G/4G).

In cellular technology, NB-IoT and LTE-MTC can be used for the IoT. These compete with LoRa and Sigfox. Their usage depends on their use cases.

For wearable health devices, we are working on Bluetooth Low Energy. Other technologies can also be used, but since BLE is built in the smartphone, the latter can work as an aggregator and send data.

Q. What do you think of security when it comes to the IoT?

A. It is a vast subject. We are working on Security by Design Principles and Trust Centre. Hardware and software security features are required. Mandatory Testing & Certification of Telecom Equipment (MTC) scheme has already given a notification on this and essential requirements are being prepared. Under it, all communicating devices will come under security testing.

Q. Is the IoT helping the ESDM sector?

A. The credit goes to semiconductor companies for their innovations. They have developed sensors, chips and more, and are miniaturising them.

Q. Will lack of hardware infrastructure affect the IoT ecosystem in India?

A. No!

Q. What is the recipe for success from a sales perspective?

A. Smart devices should also be secure because the brand value depends on that. Secure devices must be deployed before implementation of MTCTE policy. **EFY**